

Masters Program in **Geospatial Technologies**



MULTI-SENSOR INLA-SPDE BAYESIAN FLOOD MAPPING: A UNITED NATIONS CASE STUDY

Francisco J. Lozada

Dissertation submitted in partial fulfilment of the requirements
for the Degree of *Master of Science in Geospatial Technologies*

Multi-sensor INLA-SPDE Bayesian Flood Mapping: A United Nations Case Study

Francisco J. Lozada-Aguilar

 [0009-0008-2727-8260]

Supervised by:

Prof. Dr. Jorge Mateu
Universitat Jaume I
Castellon, Spain

Co-supervised by:

Prof. Dr. Benjamin Risse
Universität Münster
Münster, Germany

Prof. Dr. Mauro Castelli
Universidade NOVA de Lisboa
Lisbon, Portugal

Software Availability:

DOI [10.5281/zenodo.18442227](https://doi.org/10.5281/zenodo.18442227)



January, 2026

STATEMENT OF INTEGRITY

I hereby declare that the thesis entitled “Multi-sensor INLA-SPDE Bayesian Flood Mapping: A United Nations Case Study” represents my own original work. I confirm that I have not used any sources, data, or assistance other than those explicitly acknowledged and referenced in this thesis. All contributions from other individuals have been duly recognized, and all sources, whether published or unpublished, have been properly cited.

Furthermore, I declare that this thesis has not been previously submitted, in whole or in part, for any degree or academic qualification at this or any other institution.

Castellón de la Plana, January 2026

Francisco J. Lozada,

ACKNOWLEDGMENTS

I would like to express my sincere gratitude, first and foremost, to the supervisors of this work. I am especially grateful to **Prof. Dr. Jorge Mateu**, whose ideas and discussions were particularly illuminating for the application of probabilistic techniques that form the core of this research. I also thank **Prof. Dr. Benjamin Risse**, from whom I learned computer vision techniques, and who provided guidance and continuous support during my stay in Münster through his seminar group, where this thesis gradually took shape from a conceptual perspective. I am also very thankful to **Prof. Dr. Mauro Castelli**, who kindly offered his support in supervising the writing and the final report, a critical stage to ensure the quality and success of this work.

More broadly, I would like to extend my gratitude to the **entire academic staff of the Erasmus Mundus Master in Geospatial Technologies** at Jaume I University of Castellón, the University of Münster, and NOVA University of Lisbon, who generously share their knowledge in service of the geospatial sciences. I also wish to thank the coordinators of this program, **Prof. Dr. Joaquín Huerta** (UJI), **Prof. Dr. Christoph Brox** and **Prof. Dr. Filipe Feitosa** (University of Münster), and **Prof. Dr. Marco Painho** (NOVA IMS). Studying this master's program has been a truly once-in-a-lifetime experience.

This project was made possible thanks to the United Nations Global Service Centre. I would like to thank **Lara Prades**, from whom the initiative originated and who motivated this project, for her support and guidance in contributing to this mission. I extend my special thanks to **Silvia Menchén**, who accompanied me throughout the entire process on the technical side and provided invaluable feedback and ideas, which are strongly reflected in this work, particularly in the geospatial component and all aspects related to flood analysis. It was a great pleasure to learn from them.

I would also like to thank **Gloria Edo**, who provided continuous support throughout the program to those of us who arrived in Castellón, and whose guidance was essential in many aspects beyond the academic sphere.

My sincere thanks to all!

ABSTRACT

This thesis addresses the critical challenge of flood detection in conflict-affected regions characterized by severe limitations in ground truth data, with a specific focus on South Sudan in collaboration with the United Nations Global Service Centre. Within the broader context of the Anthropocene and the intensification of climate-driven hydrological extremes, conventional pixel-based classification approaches often fail to capture the spatial continuity of flooding processes and remain highly sensitive to noise and uncertainty inherent in satellite observations.

To address the dual challenges of small data regimes and multi-source sensor fusion, this research proposes a Bayesian spatial modeling framework based on Integrated Nested Laplace Approximation combined with Stochastic Partial Differential Equations (INLA-SPDE). By conceptualizing flood occurrence as a latent spatial process rather than a collection of independent discrete events, the model coherently integrates heterogeneous Earth Observation data (including Sentinel-1 SAR backscatter and Sentinel-2 optical spectral indices) together with physically interpretable static topographic covariates.

A harmonized dataset covering ten Areas of Interest (AOIs) in South Sudan was specifically constructed for this study, comprising 130 flood rasters used for both training and validation. The results demonstrate that the proposed INLA-SPDE framework produces robust probabilistic flood susceptibility maps with outstanding predictive performance (AUC = 0.9949). This work constitutes, to the best of our knowledge, the first implementation of a binomial INLA-SPDE model for satellite-based natural hazard detection, advancing the field from deterministic flood mapping toward principled, evidence-based probabilistic inference in support of humanitarian decision-making.

INDEX

1. Introduction	1
1.1 Context and Motivation. Floods in the context of United Nations operations	1
1.2 Why floods must be treated as (bayesian) probabilistic spatial process	1
I. Bayesian inference as epistemological framework for hydrological hazards	1
II. The INLA-SPDE approach	3
1.3 Earth Observation data as partial evidence	5
1.4 Related Work	8
1.5 Research Gap	9
I. The “small data” challenge	10
II. The “fusion” challenge	10
III. The “spatial dependency” challenge	10
1.6 Research Questions	10
1.7 Research Objective	10
1.8 Expected Contributions	10
1.9 Thesis structure	11
2. Data and Study Area	13
2.1 Dataset creation	13
I. Static data	14
II. Dynamic data (date-associated)	15
3. Methodology	19
3.1 Methodological Overview	19
3.2 Algorithm 1. Bayesian Flood Mapping via INLA-SPDE	19
3.3 Algorithm implementation. of Bayesian Spatiotemporal Flood Mapping via INLA-SPDE	22
3.4 Reproducibility and software implementation	23
4. Results and Discussion	25
4.1 Posterior Estimates of Fixed Effects	25
Latent Spatial Process (SPDE)	25
4.2 Flood Susceptibility Mapping	27
4.3 Spatiotemporal Prediction (Testing Stage)	27
5. Conclusion and Future perspectives	29
Epilogue. On the Anthropology of Climate Change	30

1. INTRODUCTION

1.1 CONTEXT AND MOTIVATION. FLOODS IN THE CONTEXT OF UNITED NATIONS OPERATIONS

In the era of the Anthropocene¹, climate change poses a global challenge for many populations, especially vulnerable groups such as people living in areas prone to natural disasters, where extreme events like droughts, wildfires, and floods occur. The last ones are the focus of this work, as 44% of natural disasters between 1970 and 2019 have been associated with flooding (Intergovernmental Panel on Climate Change (IPCC), 2023).

This study focuses on pluvial and river floods. In recent decades, their frequency and magnitude have increased due to the rise in extreme precipitation events driven by climate change. This trend is expected to continue, with high confidence, in regions such as Asia, Central Africa, the area of interest in this work, 10 areas of interest (AOI's in South Sudan), Western Europe, Central and South America, and Eastern North America (Intergovernmental Panel on Climate Change (IPCC), 2023).

South Sudan is one of the most flood-prone and vulnerable countries in the world. For instance, in the Sudd wetland region, covering approximately 90,000 km², recurrent flooding led to the displacement of an estimated 1.8 million people between 2019 and 2022; these floods severely affected communities in the states of Jonglei, Unity, and Upper Nile, inundating four of the country's ten states. These affected areas are predominantly located on extensive floodplains, where the interaction between the White Nile hydrological system, the Sudd wetland dynamics, and low topographic gradients promotes prolonged flood persistence and high exposure of human settlements (Chol et al., 2026).

For these reasons, this work focuses on developing a decision-support system through the practical implementation of geospatial technologies and modeling approaches aimed at assisting humanitarian and peacekeeping operations.

This work is being developed in collaboration with United Nations Global Service Centre, for the peacekeeping strategies and decision making on the natural disasters in 10 areas of interest South Sudan (see FIG 1 below).

1.2 WHY FLOODS MUST BE TREATED AS (BAYESIAN) PROBABILISTIC SPATIAL PROCESS

I. Bayesian inference as epistemological framework for hydrological hazards

In nature, phenomena rarely present themselves as certainties. Our observations are partial signals from noisy measurements (in this project, radar backscatter, optical spectral reflectance, and terrain properties) from which we must infer what is most the likely phenomenon represented in this sample. For this project, we adopt a Bayesian epistemological framework not as a matter of preference, but as a consequence of the nature of the problem itself, floods as not directly observed entities (as we mentioned in last section, this project was developed for United Nations Global Service Centre, for remotely early detection of this natural disaster events and risk assessment in vulnerable zones).

¹ In this work, the term Anthropocene is not simply a diffuse by-product of "human activity" in the abstract. Rather than treating humanity-as-a-whole as a single geological agent, the concept has been criticized for obscuring heterogeneity among humans, differences in ontologies, responsibilities, and modes of world-making. Many accounts locate its onset in the eighteenth century CE, closely tied to the Industrial Revolution and the expansion of fossil-fuel-based production (Schaffhauser, 2022).

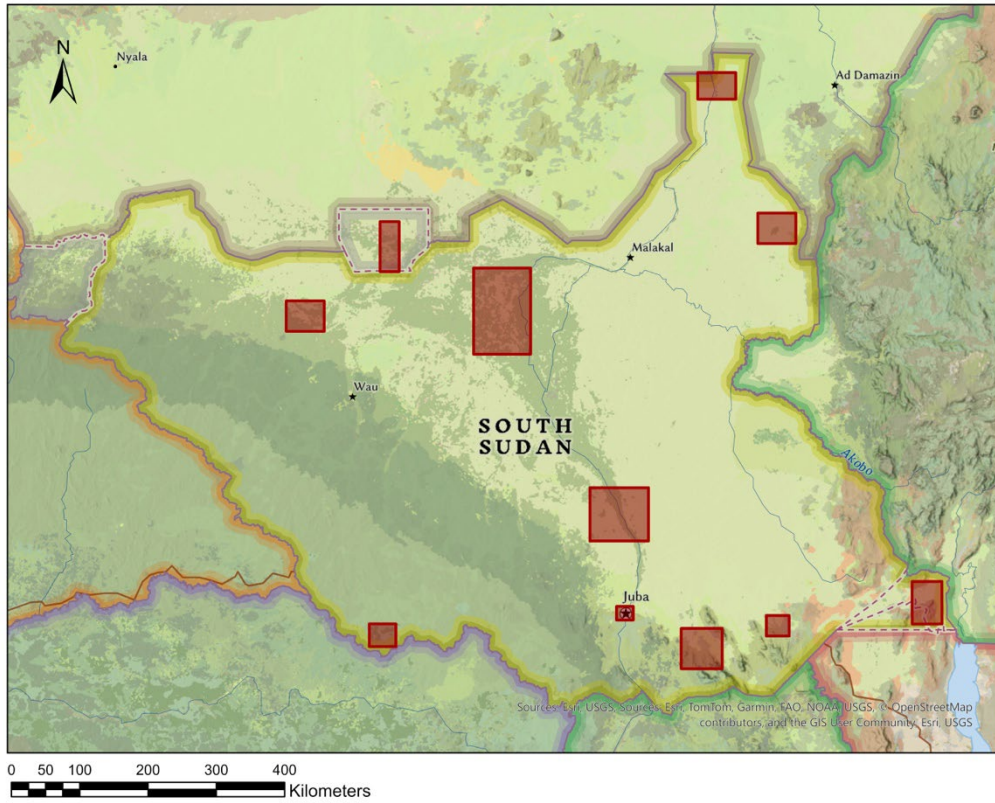


FIG 1. Areas of Interest (AOI's) in South Sudan. Source: Author's own elaboration

At any given spatial location, the problem can be formulated as two mutually exclusive and exhaustive hypotheses: either a flood is present, or it is not. However, instead of asking the forward question, “What data would a flood produce?”, Bayesian inference allows us to reverse the problem in a more meaningful way: given the data we observe, what is the probability that a flood is occurring?

This inversion is crucial in low-data scenarios. As will be discussed in the Research Gap section, one of the main challenges in Central African flood monitoring, the study area of this work, is the limited availability of labeled observations (in the spatial domain, these labels must have a reference to a specific location on Earth and, since there are images, each pixel represents an area, and if this area is large, or, in other words, the image resolution is low, the location loses precision). Under those conditions, approaches such as convolutional neural networks, transformers, or other deep learning architectures become impractical because these algorithms require large volumes of data.

Bayesian inference offers a natural mechanism to overcome this limitation. Prior probabilities encode physically and hydrologically plausible expectations, which can be continuously updated as new observations become available. Importantly, these priors are not static assumptions, but dynamic components of an inferential process that evolves with evidence.

Yet floods cannot be adequately described as independent pixel-level events. Hydrological processes exhibit strong spatial continuity: water accumulates, and propagates across landscapes following terrain, soil properties, and connectivity. Treating floods as isolated pixels ignores the physics that governs their behavior.

To address this, we model flooding as a latent spatial field, represented by a Gaussian random field that captures the underlying spatial dependence of the process. This approach allows information to propagate across space, enabling inference even in areas with limited or noisy observations (something usual in this field of study, satellite data for flood detection). Unlike traditional machine learning techniques such as

clustering or principal component analysis, which primarily capture statistical similarity, the spatial Bayesian framework encodes spatial causality and continuity.

In general, our main proposal is studying floods not as discrete objects, but as spatially varying fields whose magnitude and extent emerge probabilistically from the interaction between data, prior knowledge, and physical constraints. Bayesian spatial inference therefore provides a framework that is not only statistically robust, but also conceptually aligned with the hydrological nature of flooding phenomena. For this aim, we will use the INLA-SPDE method as the inference algorithm.

II. The INLA-SPDE approach

Once the main advantages of a Bayesian framework and the challenges posed by spatial dependence have been established, we can propose an approach to implement this epistemological analysis by focusing on two core components: the *inferential engine* and the *spatial representation of the phenomenon*.

For this work, Integrated Nested Laplace Approximation (INLA) was selected as the inference framework for estimating posterior probabilities. The main reason to choose INLA framework, a deterministic approach, over Markov chain Monte Carlo (MCMC), a widely used stochastic simulation method, lies on the trade-off between convergence, efficiency, and the nature of the resulting error.

MCMC shows better error estimation: in general, error can be as small as possible, condition on the computational time the simulation runs. Error in MCMC, decreases at rate $\mathcal{O}\left(N^{-\frac{1}{2}}\right)$, where N denotes the number of samples generated by the algorithm. This means that reducing the error by one order of magnitude requires increasing the sample size by two orders of magnitude, i.e., $N \rightarrow \text{Error}$, $100N \rightarrow \frac{\text{Error}}{10}$, $1000N \rightarrow \frac{\text{Error}}{100}$, ... (Rue et al., 2009), in general

$$\text{Error}_N = \left(N^{-\frac{1}{2}}\right) \Rightarrow \lim_{N \rightarrow \infty} \text{Error}_N = 0 \quad (1)$$

Therefore, although the error theoretically converges to zero as $N \rightarrow \infty$ (called asymptotic optimality), in practice this convergence is often prohibitively expensive in terms of computational time, particularly for latent Gaussian models with high-dimensional structures. As discussed in (Rue et al., 2009), the overall complexity of MCMC in such stings can scale cubically ($\mathcal{O}(n^3)$) with the number of latent variables.

Given that this thesis is focused on practical implementation for humanitarian decision-making, an alternative approach is required, one that prioritizes timely and stable inference. INLA addresses this requirement by replacing stochastic simulation with deterministic approximations. It is said to be deterministic because rather than relying on random samples, INLA operates on the full dataset and approximates posterior marginal distributions through numerical integration, a set of sums defined in series, in this case, Laplace approximations, which computationally utilize Taylor series expansions (Rue et al., 2009).

In INLA framework, uncertainty does not arise from sampling variability, but from approximation itself, i.e., error introduced by INLA is structural rather than stochastic: while MCMC errors are additive and random, INLA errors are systematic and, therefore, more predictable.

On the other hand, from a computational perspective, the overall complexity of the algorithm can be reduced to approximately $\mathcal{O}\left(n^{\frac{3}{2}}\right)$ as long as the number of hyperparameters is relatively small, an assumption satisfied by the models considered in this work. Here, n denotes the number of nodes in the two-dimensional triangular mesh constructed over the study area, a nested dissection of the map (this is why, the algorithm is called Integrated Nested Laplace Approximation; the construction of this mesh is described in the following paragraph). Hereby, instead of treating each pixel as directly dependent on all others, an

approach that would require dense matrix factorization with complexity $\mathcal{O}(n^3)$, the spatial domain is discretized into n mesh nodes connected locally through triangles. Each node is assumed to interact only with its immediate neighbors, while interactions with distant nodes are set to zero. This local dependency structure leads to a “sparse” matrix and allows the spatial field to be modeled as a Gaussian Markov Random Field (GMRF), which is central to the proposed approach (Lindgren et al., 2011).

Hence, the Integrated Nested Laplace Approximation (INLA) as formally introduced by Rue et al., (2009) provides a computationally efficient, deterministic framework for approximate Bayesian inference to latent Gaussian models with analytic approximations combined with low-dimensional numerical integration. INLA achieves this by constructing nested approximations to compute the posterior marginals of the latent variables x_i , defined mathematically as

$$\tilde{\pi}(x_i|y) = \int \tilde{\pi}(x_i|\theta, y) \tilde{\pi}(\theta|y) d\theta \quad (2)$$

To resolve this integral, the method approximates the marginal posterior of the hyperparameters θ using a Laplace approximation, which is expressed as

$$\tilde{\pi}(\theta|y) \propto \frac{\pi(X, \theta, y)}{\tilde{\pi}_G(X|\theta, y)} \Big|_{X=X^*(\theta)} \quad (3)$$

where $\tilde{\pi}_G(X|\theta, y)$ represents the Gaussian approximation to the full conditional distribution of the latent field X , and $X^*(\theta)$ is its modal configuration for a given hyperparameter vector θ .

By exploiting the sparse precision matrices inherent to Gaussian Markov Random Fields (GMRF), INLA systematically evaluates these equations to integrate out the hyperparameters θ , yielding highly precise posterior estimates at a fraction of the computational cost of traditional MCMC methods while preserving strict mathematical rigor.

In the context of the INLA framework, Lindgren et al., (2011) showed that, to handle the *big-n* problem described above, Matérn covariance is recontextualized to overcome the computational bottleneck of dense spatial matrices. Formally, Matérn covariance function between two spatial locations $u, v \in \mathbb{R}^d$ is defined as

$$r(u, v) = \frac{\sigma^2}{2^{v-1}\Gamma(v)} (\kappa|v - u|)^v K_v(\kappa|v - u|) \quad (4)$$

where K_v is the Bessel function, σ^2 is the marginal variance, $v > 0$ is the mean-square differentiability (smoothness) of the field, and $\kappa > 0$ is a scaling parameter controlling the practical correlation range $\rho \approx \frac{\sqrt{8v}}{\kappa}$.

Intuitively, instead of modeling spatial dependence by calculating this complex distance-based correlation between every single pair of points in a landscape, the SPDE approach conceptualizes the continuous Matérn field as a physical medium reacting to random, independent local shocks. Rigorously, a Gaussian Field (GF) exhibiting a Matérn covariance is exactly the stationary solution to the linear fractional stochastic partial differential equation (SPDE) expressed in equation (5) below.

On the other hand, to represent spatial dependence in flood occurrence, we adopt the Stochastic Partial Differential Equation (SPDE) framework as a unifying approach. This formulation is particularly well suited for the present project, as it enables a spatially continuous latent process. Here, flood occurrence is

modeled through a Matérn Gaussian random field to be efficiently approximated by a discrete representation defined over the triangular mesh defined above. The mesh is constructed over the spatial domain using the locations of point pattern observations, which in this case correspond to binary outcomes (flooded versus non-flooded pixels).

According to (Lindgren et al., 2011), the SPDE defining the latent spatial field is given by

$$(\kappa^2 - \Delta)^{\frac{\alpha}{2}} x(\mathbf{u}) = W(\mathbf{u}) \quad (5)$$

$$\mathbf{u} \in \mathbb{R}^2, \quad \alpha = \nu + \frac{d}{2}, \quad \kappa > 0, \quad \nu > 0$$

The intuition behind the definition in terms of flooding is as follows. Let $\mathbf{u}$ denote a pixel observed by a satellite over a given territory, and let $x(\mathbf{u})$ represent the latent propensity of that pixel to be flooded, conditioned on unobserved environmental processes (flood observations are recorded at discrete pixel locations, but $x(\mathbf{u})$, the underlying physical process is inherently continuous in space and therefore must be modeled as a smooth spatial field).

Now, the term $W(\mathbf{u})$ represents spatially uncorrelated Gaussian white noise and captures variability that cannot be explained by observed covariates at the time of flooding (in this work, satellite data, elevation, vegetation, etc.). In contrast, the differential operator $(\kappa^2 - \Delta)^{\frac{\alpha}{2}}$ in equation (5) governs how this unstructured noise is transformed into a spatially correlated field with Matérn covariance structure. The Laplacian operator Δ in equation (6), enforces local spatial interaction by penalizing abrupt changes between neighboring locations, thereby controlling the smoothness of the field (Lindgren et al., 2011),

$$\Delta = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} \quad (6)$$

Finally, solutions to the SPDE are not obtained in closed form in the spatial domain. Instead, the fractional differential operator is interpreted in the spectral domain via the Fourier transform, where it becomes a simple multiplication (Lindgren et al., 2011). This representation yields an explicit Matérn covariance structure and allows for an efficient numerical approximation through finite element methods defined on the triangular mesh.

1.3 EARTH OBSERVATION DATA AS PARTIAL EVIDENCE

The variables introduced in the previous section are driven by satellite observations, which constitute the core input of the proposed modeling framework. These observations are obtained from two distinct physical sensing mechanisms, each capturing different properties of the Earth's surface (for further details, see the Data and Study Area section).

This section briefly describes the nature of these observations (in this work, the observations are based on Copernicus satellites, Sentinel-1 and Sentinel-2), the underlying physical principles governing the sensing process, and the specific sources of uncertainty that each introduces into the model. Such uncertainties arise not only from sensor limitations and atmospheric effects, but also from the indirect nature of remote sensing, where surface properties are inferred rather than directly observed.

I. Sentinel-2: Optical Sensors

In this type of sensing systems, imagery depends on solar illumination; Incoming solar radiation interacts with Earth surface and is reflected back to the sensor. For flood detection, open water exhibits strong absorption of electromagnetic radiation, especially the Near-Infrared (NIR) and shortwave infrared

(SWIR) region of the spectrum, therefore, flooding in optical imagery manifests as a disappearance or strong attenuation of the reflected signal (see FIG 2 below). Despite this conceptual simplicity, optical flood detection suffers from a fundamental limitation: cloud cover. Since optical sensors rely on solar radiation reaching the surface, the presence of clouds severely degrades or entirely prevents flood observation, particularly during extreme weather events when floods are most likely to occur (Jong & Meer, 2004).

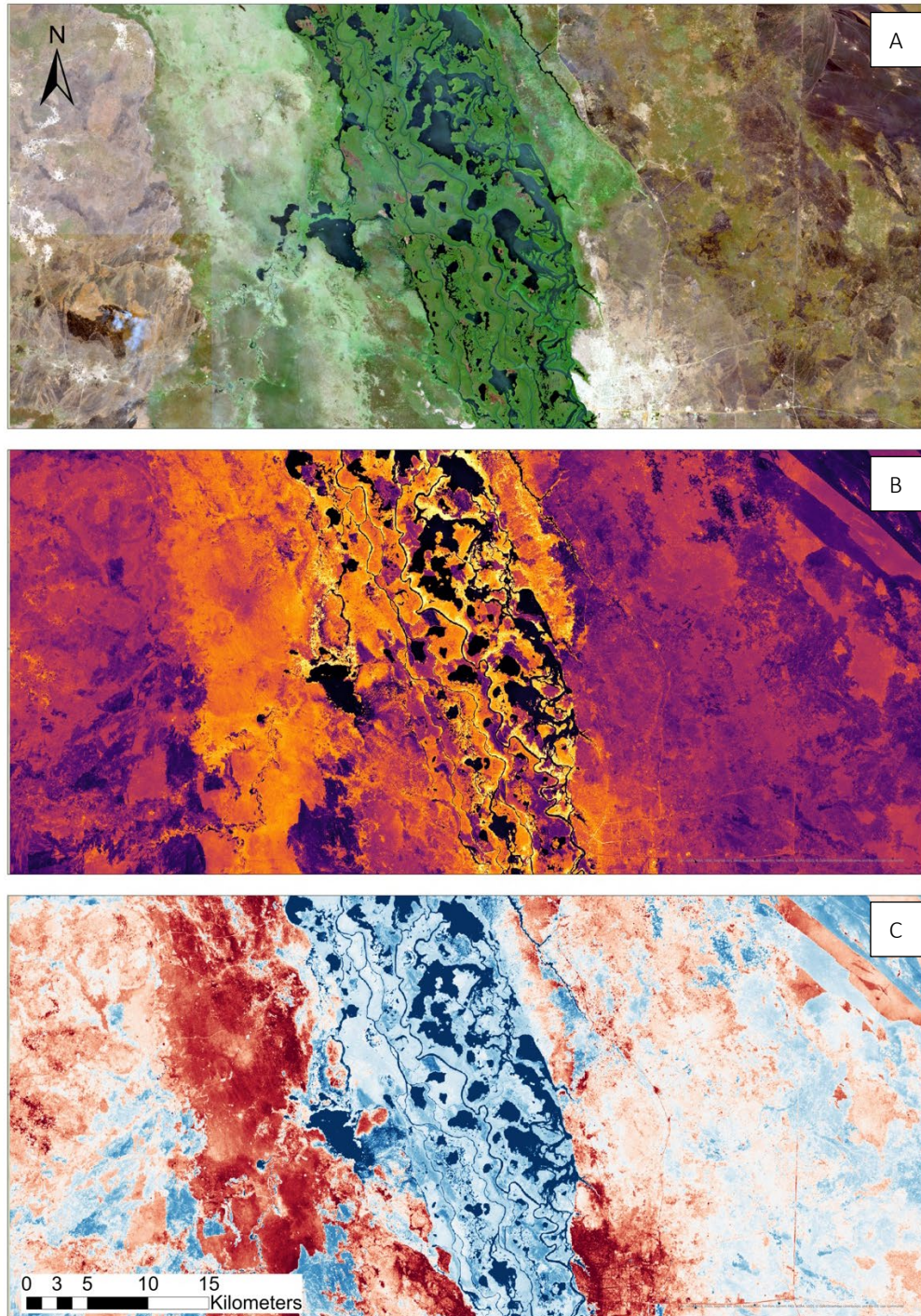


FIG 2. (A) True color image with water bodies; (B) Water detection with optical NIR Band; (C) Water detection with optical SWIR Band. Satellite data downloaded from Sentinel-2 Copernicus Browser. Source: Author's own elaboration

II. Sentinel-1: Syntetic Aperture Radar

To overcome atmospheric limitations that optical sensors induce, flood detection often relies on active microwave sensors, particularly Synthetic Aperture Radar (SAR). Unlike optical systems, SAR instruments emit their own electromagnetic signal and measure the backscattered echo, enabling observations independent of daylight and cloud cover (Stephens et al., 2012). The key physical principle governing SAR-based flood detection is specular reflection, when an incident radar pulse strikes such a surface at an oblique angle, the majority of the energy is reflected away from the sensor rather than back toward it. Consequently, flooded areas typically appear as regions of low backscatter (dark pixels, see FIG 3 below).

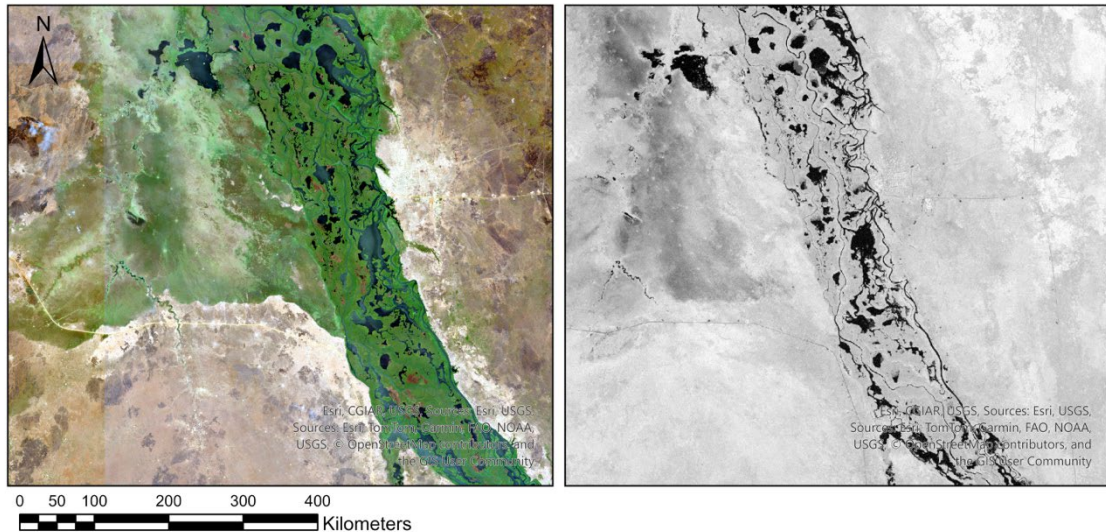


FIG 3. Water detection with SAR Sentinel-1. Satellite data downloaded from Copernicus Browser. Source: Author's own elaboration

However, as emphasized by (Stephens et al., 2012), the assumption of water surfaces as perfect specular reflectors frequently breaks down in real-world conditions. Wind-induced surface roughness disrupts the smoothness of the water surface, increasing radar backscatter and causing flooded areas to resemble rough land surfaces. In addition, the presence of emergent vegetation or built structures within inundated areas can generate double-bounce scattering mechanisms, whereby the radar signal reflects off the water surface, then off vertical structures (e.g., tree trunks), and finally returns to the sensor. This effect can cause flooded areas to appear anomalously bright rather than dark.

Following (Stephens et al., 2012), we adopt the premise that SAR observations constitute partial and noisy evidence of flooding, subject to spatially structured errors. Physically, these errors arise because the idealization of water as a specular reflector is frequently violated: wind-induced surface roughness increases backscatter, and emergent vegetation generates double-bounce effects that artificially enhance the radar return from inundated areas.

These phenomena affect contiguous spatial regions, producing clusters of uncertainty that violate the classical assumption of independent errors. To mitigate the bias introduced by such spatially correlated errors in standard models (e.g., spurious attribution to surface roughness, friction, or topography), and to enable a probabilistic assessment of flood risk that explicitly acknowledges the inherent noisiness of satellite observations, we employ the INLA-SPDE framework.

This approach explicitly models spatial dependence through a latent Gaussian field, allowing the separation of structured environmental variability (the true physical signal of flooding) from sensor- and environment-induced noise (e.g., wind and vegetation effects). The result is a set of robust probabilistic flood

susceptibility maps that remain stable in the presence of localized artifacts and systematically structured uncertainty.

1.4 RELATED WORK

Flood monitoring using remote sensing has evolved significantly, transitioning from threshold-based methods to deep learning architectures and probabilistic approaches as shown in this section. Recent literature reveals persistent limitations related to the dependence on large training datasets, and lack of explicit spatial modeling. The predominant approaches and their limitations are categorized below in the context of this study.

- **Machine Learning and Deep Learning Approaches (CNNs)**

A substantial portion of recent literature has focused on the use of Convolutional Neural Networks (CNNs) and classical machine learning algorithms. Drakonakis et al., (2022) implemented CNN-based models achieving Sentinel-1 and Sentinel-2 data fusion; however, their approach does not explicitly model spatial dependence, treating predictions as independent segmentation outputs.

Similarly, Bereczky et al., (2022) applied CNNs exclusively to Sentinel-1 data, thereby failing to leverage the complementary information provided by multi-sensor fusion.

Advanced architectures such as the ResUNet (e.g, Chen & Bruzzone, (2022)) have enabled change detection and bi-temporal fusion. Nevertheless, these methods operate primarily at the pixel level and face difficulties handling long temporal sequences or classifying complex changes due to inherent uncertainty and variability.

Zandsalimi et al., (2025) demonstrated that integrating Digital Elevation Models (DEM) with U-Net architectures improves the extraction of spatial patterns in urban environments. However, these models suffer from limited generalization capability, requiring extensive training datasets to adapt to new geographic and topographic conditions.

Traditional supervised approaches also exhibit limitations. Meyer et al., (2022) applied Support Vector Machines (SVM) and Random Forest (RF), highlighting that pixel-based classification introduces significant uncertainty in heterogeneous land cover environments. Loukili et al., (2025) further emphasized that hybrid classifiers such as RF require large labeled datasets, which represent a critical constraint in regions with limited ground truth availability.

- **Probabilistic and Bayesian approaches**

To address uncertainty, several studies have adopted probabilistic frameworks. Bauer-Marschallinger et al., (2022) proposed a Bayesian inference-based algorithm for Sentinel-1 flood detection using harmonic time-series analysis to derive a universal flood signature. While computationally efficient and capable of quantifying uncertainty, this method fundamentally relies on pixel-wise temporal analysis and does not explicitly model spatial dependence between neighboring pixels.

More recently, Ta et al., (2024) improved probabilistic estimation by combining Pareto normalization with Bayesian inference on Sentinel-1 time series, enabling the identification of flood stages and dynamic temporal characteristics. However, similar to Bauer-Marschallinger et al., (2022), this approach assumes independence between spatial observations.

Other approaches, such as Hansen et al., (2025), refined flood extent estimation using fuzzy logic and hydrological contextualization, while Kridalukmana et al., (2023) explored fuzzy Bayesian networks for early warning systems, although not specifically focused on satellite-based flood mapping.

- **Previous Studies on Flood Monitoring and Modeling in South Sudan**

Given South Sudan's hydroclimatic vulnerability and the complexity of the Sudd wetland system, several recent studies have addressed flood detection and prediction in the region. These efforts have primarily focused on overcoming the lack of in-situ observations through the use of Earth observation technologies and hydrodynamic modeling.

In the context of multi-sensor fusion for rapid impact assessment, Cian et al., (2024) developed an approach integrating radar data (Sentinel-1), optical imagery (Sentinel-2), and high-resolution satellite data (PlanetScope) to analyze the 2020 seasonal floods. Their methodology introduced a mapping algorithm based on multitemporal statistical analysis of the Modified Normalized Difference Water Index (MNDWI) derived from Sentinel-2 imagery. The results demonstrated that the exclusive use of SAR or optical sensors significantly underestimated the spatial extent of flooding, detecting only 24% and 78%, respectively, of the impact estimated by the multi-sensor approach, although with limitations in densely vegetated regions. Specifically, the core statistical estimation was performed independently for each pixel by computing temporal summary statistics, including the median and percentile thresholds, without explicitly incorporating information from neighboring pixels during the index generation process. Consequently, the flood estimation process does not account for terrain-driven spatial continuity, hydrological connectivity, or spatial uncertainty propagation.

To address the limitations of optical and SAR sensors in areas characterized by dense vegetation and persistent cloud cover, Downs et al., (2023) evaluated the performance of Global Navigation Satellite System reflectometry (GNSS-R) using the CYGNSS satellite constellation over the Sudd wetland during the October 2021 flood event. When compared with operational flood products derived from VIIRS, MODIS, and Sentinel-1, the GNSS-R approach detected 35.4% more surface water than Sentinel-1, while optical products severely underestimated flood extent, by up to 83.7% in the case of MODIS, due to signal obstruction caused by clouds and vegetation. This study highlights the potential of L-band reflected signals to penetrate vegetation and improve flood detection in wetlands such as the Sudd, although challenges remain regarding spatial resolution and sensitivity to soil moisture variability.

The literature review reveals a clear methodological gap. While deep learning approaches ((Konapala et al., 2021); (Zandsalimi et al., 2025)) require large training datasets that are often unavailable in data-scarce regions such as South Sudan, existing Bayesian methods ((Bauer-Marschallinger et al., 2022); (Ta et al., 2024)) operate under pixel-wise independence assumptions.

On the other hand, despite the advances in South Sudan as study area ((Downs et al., 2023); (Cian et al., 2024)), important limitations remain in large the wetlands of the Sudd, where hydrological connectivity and terrain flow dynamics and spatial analysis play a critical role.

Flooding is inherently a spatially continuous phenomenon characterized by strong spatial autocorrelation governed by topography, hydrological connectivity, and terrain structure (see research gap in following section).

1.5 RESEARCH GAP

Given the background explained above, research gap can be grouped into three main categories: data, multi-modal fusion and spatial dependency. Bayesian frameworks have been extensively developed and successfully applied in modeling spatial and spatiotemporal phenomena, but no studies have yet applied INLA-SPDE directly to flood detection using remote sensing data (in general, for natural hazard detection with multi-source remote sensing techniques). Existing research on INLA-SPDE in natural disasters remains largely theoretical or focused on other phenomena, without concrete applications to monitoring natural hazards such as floods with real satellite data (remote sensing and Earth Observation).

The current literature reveals three main gaps:

I. The “small data” challenge

The limited availability of ground-truth flood observations in conflict zones (such South Sudan, our study area) significantly constrain model calibration, leading to reduced predictive accuracy and weak generalization across heterogeneous regions and flood regimes.

Neural network approaches (e.g., CNNs, Transformers, and U-Net) require large volumes of data, which makes gradient optimization difficult. In other words, it is challenging to achieve convergence toward a general solution that remains valid across multiple scenarios, such as different seasonal conditions, varying illumination, and the presence of confusing reflective surfaces (e.g., sandbanks or recently irrigated areas).

II. The “fusion” challenge

Limited integration with multi-source remote sensing. Despite the increasing availability of multi-temporal Synthetic Aperture Radar (SAR) and optical data (e.g., Sentinel-1 and Sentinel-2), no studies have coupled INLA-SPDE with these datasets for flood modeling. Fusing Satellite Optical data (spectral reflectance) and Satellite SAR data (backscatter and texture) is non-trivial. They belong to distinct feature spaces with different distributions, and different physical interpretation.

III. The “spatial dependency” challenge

Flood events involve complex spatial heterogeneity and temporal dynamics, yet existing frameworks rarely account for these non-stationary and anisotropic structures. Floods are spatially continuous phenomena, standard pixel-based classification assume data is independent and identically distributed, ignoring the strong spatial autocorrelation of these phenomena

As a result, there is currently no established methodological framework that employs INLA-SPDE as the core probabilistic model for flood detection from satellite imagery. In general, bridging this gap offers an opportunity to extend spatial binomial modeling into the domain of hazard assessment, creating a statistically grounded approach to hazards monitoring and prediction (in this study, flooding).

1.6 RESEARCH QUESTIONS

RQ1. How does the integration of Sentinel-1 SAR features and Sentinel-2 optical covariates influence the inference and spatial prediction of flood occurrence within a Bayesian INLA-SPDE framework?

RQ2. Under which modeling and discretization conditions (mesh resolution, prior specification of spatial range and variance, and event prevalence) does the INLA-SPDE approach yield stable, well-calibrated posterior inference and accurate spatial predictions in flood detection applications?

1.7 RESEARCH OBJECTIVE

To develop and analyze a Bayesian spatial modeling framework based on INLA-SPDE for flood occurrence detection using multi-sensor satellite data, and to characterize the behavior of the latent spatial field and posterior predictions under different modeling and discretization conditions.

1.8 EXPECTED CONTRIBUTIONS

This thesis directly addresses the research gap identified in Section 1.3 by developing, to the best of our knowledge, the first implementation of a binomial INLA–SPDE–based framework for satellite-driven natural hazards detection (in this work, floods), explicitly integrating a covariate-based model with multi-sensor Earth Observation data from Sentinel-1 SAR and Sentinel-2 multispectral imagery.

Unlike existing applications of INLA-SPDE in environmental sciences, which primarily focus on the interpolation or smoothing of continuous geophysical variables, this work extends the framework to discrete hazard detection, treating satellite observations as noisy measurements of an underlying latent flood-risk process. The main contributions of this thesis are summarized as follows:

Contribution 1: Dataset creation. For this work, we created the first flood dataset for South Sudan, covering 10 areas of interest (see FIG 1). The dataset includes flood extent labels, Sentinel-1 data, and Sentinel-2 derived spectral indices, comprising 19 flood events for training and 3 flood events for testing, between October 2019 and August 2025.

Contribution 2: Bayesian framework for probabilistic flood detection. We propose a Bayesian framework based on multi-sensor satellite imagery (Sentinel-1 and Sentinel-2), in which satellite observations are explicitly treated as noisy measurements of a latent hydrological risk process. This framework enables learning from an initial dataset and subsequently updating prior beliefs through posterior inference as new data become available, thereby directly addressing the “Small Data Challenge” presented in previous section.

Contribution 3. Integration of heterogeneous environmental covariates derived from Earth Observation data (e.g., MNDWI, NDVI, DEM, slope, curvature), enabling a physically interpretable representation of flood dynamics within a Bayesian spatial modeling paradigm, addressing the “Fusion Challenge” presented in previous section.

Contribution 4. Definition of a probabilistic modeling paradigm for natural disaster detection, reframing satellite imagery from deterministic map generation to evidence-based inference of latent environmental risk fields, addressing the “Spatial Dependency Challenge” presented in previous section

This thesis presents the first spatial Bayesian framework based on INLA-SPDE for the probabilistic detection of floods from multi-sensor satellite imagery (Sentinel-1 and Sentinel-2), in which satellite observations are explicitly treated as noisy measurements of a latent hydrological risk process.

Unlike previous work that primarily uses INLA-SPDE for the interpolation of continuous environmental variables or as a spatial smoothing tool, this work extends the approach to discrete natural hazard phenomena, integrating pixel-level remote sensing data, explicit quantification of uncertainty, and a decision-making approach for disaster management.

1.9 THESIS STRUCTURE

This document is structured into seven main chapters that guide the reader from the conceptualization of the problem to the technical implementation and final reflections. The content of each section is outlined as follows.

Introduction. This chapter establishes the context and motivation of the study, highlighting the challenges posed by flooding events in United Nations operations and the need to address these phenomena as probabilistic spatial processes. The Bayesian epistemological framework is introduced, key research challenges are defined (such as data scarcity and sensor fusion), and the research questions, objectives, and expected contributions of the thesis are clearly articulated.

Data and Study Area. This chapter describes the geomorphological and hydrological characteristics of South Sudan and the Sudd wetland region. It details the dataset construction process, explaining the generation of static covariates (such as the digital elevation model and its derivatives) and dynamic covariates (Sentinel-1 and Sentinel-2 satellite imagery), as well as the construction of ground truth based on historical flood detections.

Methodology. This chapter presents the proposed methodological framework, detailing the Bayesian flood mapping algorithm based on the INLA-SPDE approach. It explains the construction of the triangular mesh for spatial domain discretization, the formulation of the Stochastic Partial Differential Equation

(SPDE) to model the latent Gaussian field, and the implementation of the software in the R programming language for spatial inference and prediction.

Results and Discussion. This chapter analyzes the posterior estimates of the fixed effects (covariates) and visualizes the latent spatial process. The predictive performance of the model is evaluated using metrics such as the ROC curve and the Area Under the Curve (AUC). Resulting flood susceptibility maps are presented, and the model's ability to generalize to unseen test dates is discussed.

Conclusions and Future Perspectives. This chapter summarizes the main findings in relation to the research objectives and outlines future research directions, including the development of spatio-temporal models and the integration of new artificial intelligence architectures.

2. DATA AND STUDY AREA

The study areas are ten Areas of Interest (AOIs) in South Sudan and Abyei, East Africa (see FIG 1), proposed by United Nations. The main physical and geomorphological properties identified are as follows. The Sudd wetland is described as a shallow depression. This feature is fundamental to its hydrological function, as the overall shape of South Sudan contains a central lowland that acts as an accumulation conveyor for floodwaters. This geomorphological configuration facilitates widespread water retention and seasonal inundation. The region exhibits a very slight northward inclination, with an average gradient of only 1.0 to 1.5 cm per kilometer. This minimal slope significantly reduces drainage efficiency, promoting prolonged water residence times and the development of extensive swampy areas (Chol et al., 2026).

These geomorphological properties cause water originating from upstream basins to drain into and persist within this depression. As a result, the Sudd maintains year-round hydrological connectivity and functions as the largest tropical wetland in the world, with profound implications for flood dynamics, ecosystem services, and human vulnerability (Chol et al., 2026).

To capture these topographic controls within the flood detection model, a Digital Elevation Model (DEM) was incorporated as a key explanatory variable. The DEM was obtained from the Copernicus Browser (see FIG 4). The three-dimensional representation of South Sudan reveals a terrain characterized by very low elevation variability, with broad, gently sloping plains and an absence of pronounced relief.

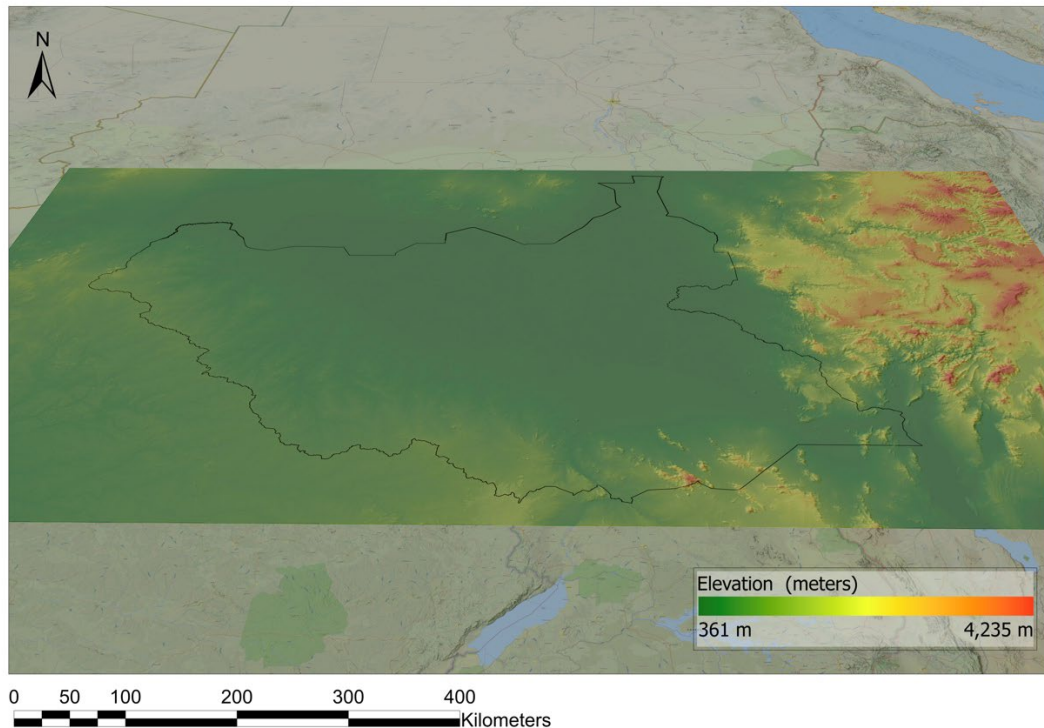


FIG 4. South Sudan Digital Elevation Model (DEM; inside the polygon). Data downloaded from Copernicus Browser.
Source: Author's own elaboration

2.1 DATASET CREATION

The dataset used to train the flood detection model is composed of two complementary categories of data: static and dynamic covariates. Together, static and dynamic data constitute the set of model covariates,

that is, the observed variables describing the environmental conditions under which flooding may occur. Static covariates encode long-term spatial predispositions to inundation, while dynamic covariates represent short-term forcing and observable responses. The following sections describe in detail the specific elements included in each category of covariates.

I. Static data

Static data represent the intrinsic and time-invariant characteristics of the territory. These variables are derived from the Digital Elevation Model (DEM) and related topographic descriptors, such as elevation and terrain morphology. Because they are physically attached to the landscape, their values remain constant throughout the observation period and provide a structural constraint on flood occurrence.

- **South Sudan Digital Elevation Model (DEM)** shown in FIG 4
- **Slope.** Slope was derived from the DEM using ArcGIS Pro, following Esri's differential terrain formulation. In this approach, slope is defined as the magnitude of the gradient of the elevation surface, computed from the partial derivatives of elevation in the x and y directions (equation 7)

$$Slope = \tanh^{-1} \left(\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \right) \quad (7)$$

where $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ represent the rate of elevation change in the east–west and north–south directions, respectively, estimated from neighboring DEM cells (Esri, n.d.-b).

- **Curvature.** Curvature was derived from DEM by computing the second-order derivatives of the surface on a cell-by-cell basis. The curvature corresponds to the negative sum of these second derivatives (equation 8), scaled by a constant factor, and represents the rate of change of slope across the surface. This metric provides information on terrain concavity and convexity, which is relevant for characterizing surface processes such as flow acceleration, erosion, and sediment accumulation (Esri, n.d.-a).

$$curvature = -100 * \left(\frac{d^2 z}{dx^2} \right) + \left(\frac{d^2 z}{dy^2} \right) \quad (8)$$

- **Heigh Nearest Above Drainage (HAND).** This model represents a normalized topological framework that quantifies the vertical distance of any point on the terrain relative to its associated drainage channel, thereby isolating local topographic gradients from absolute elevation. By defining the local draining potential, or the relative gravitational potential energy of soil water, HAND serves as a robust proxy for the depth of the water table, effectively delineating distinct soil water environments and hydrological classes across complex landscapes (Nobre et al., 2011).
- **Topographic Wetness Index (TWI).** This index is a physically based attribute designed to quantify the topographic control on hydrological processes, specifically modeling the spatial distribution of soil moisture and zones of surface saturation. Formulated as the natural logarithm of the

ratio between the specific catchment area and the local slope gradient, the index identifies areas of potential water accumulation by assuming that locations with identical TWI values exhibit similar hydrological responses to rainfall events (Qin et al., 2011).

II. Dynamic data (date-associated)

In contrast, dynamic data correspond to observations explicitly associated with a specific point in time. This category includes satellite-based measurements acquired at different dates, such as optical and SAR imagery, as well as time-referenced ancillary information (e.g., field photographs or reports) that may indicate the presence of a flood event. These variables capture the temporal evolution of surface conditions and hydrological states. The construction of the dynamic dataset was performed in a stepwise manner, in order to ensure temporal consistency between observations and to preserve the physical meaning of each covariate. The following subsections describe each step of this process in detail.

• STEP 1. The current UNOSAT Product

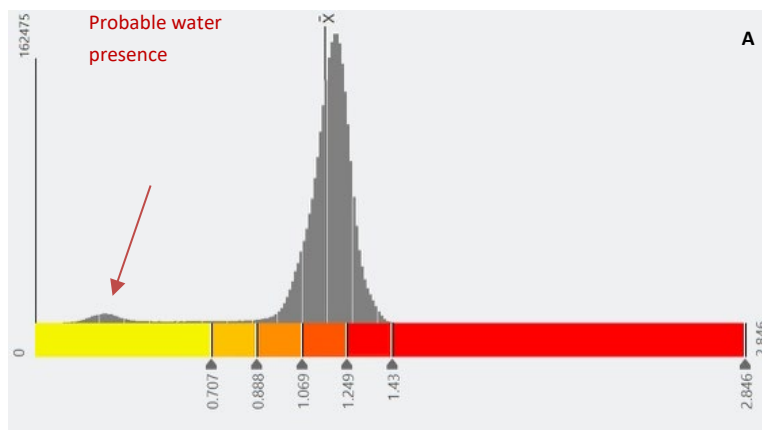
The first step consists of flood extent polygons detected by the VIIRS sensor and associated with specific event dates, which were obtained from the OCHA HDX platform. A key limitation of this source is the low spatial resolution of VIIRS (375 m per pixel), which restricts its ability to capture fine-scale inundation patterns. Using the flood detection dates provided by VIIRS, we subsequently acquired twelve times higher-resolution satellite imagery from the Copernicus program, specifically Sentinel-1 (SAR) and Sentinel-2 (optical), 30 m per pixel, corresponding to the dates on which flooding was reported by VIIRS.

These OCHA HDX polygons are used as reference (ground truth) data for flood delineation and analysis (it should be noted that the OCHA HDX platform provides only flood extent polygons and does not include the original satellite imagery)

• STEP 2. SAR Satellite Data

As explained in the introduction, Sentinel-1 SAR data, water appears dark because of how its dielectric and geometric properties control microwave scattering. Liquid water has a high dielectric constant, which means it strongly interacts with the incoming radar wave. Since SAR measures backscattered power, very little signal returns when there is water, hence, the surface looks dark (see FIG 3). We leverage this property in our analysis

By performing a principal component analysis (PCA) with a single component on the two SAR polarizations (vertical and horizontal), and subsequently segmenting the results based on standard deviation, we can identify pixels that are likely to correspond to water when their values fall two standard deviations below the mean (see FIG 5 below).



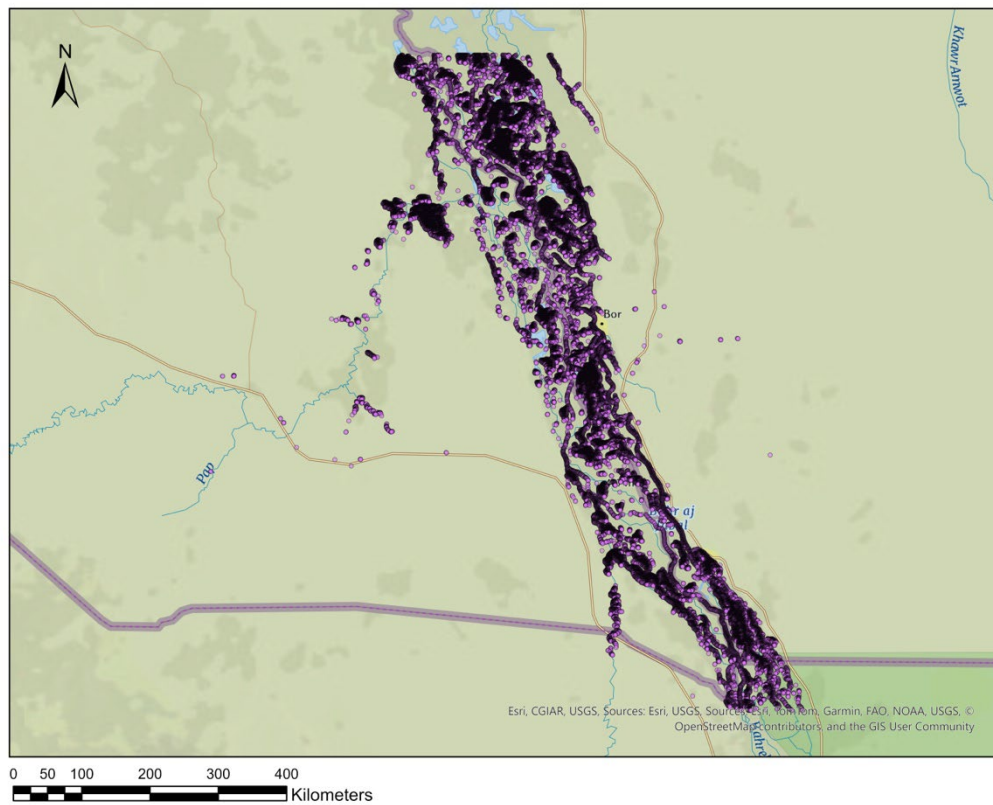


FIG 6. Flood point pattern derived from satellite observations. Source: Author's own elaboration

- **STEP 4. Spectral indices derived from Sentinel-2 bands**

The following Sentinel-2 spectral bands and derived indices were included based on their physical and hydrological properties. Water-related indices such as AWEI and MNDWI are expected to contribute positively to the probability of flooding, as higher values are associated with increased surface water presence and reduced soil infiltration capacity. In contrast, NDVI is expected to contribute negatively, since higher vegetation density enhances interception and infiltration processes, thereby reducing surface runoff and, consequently, flood susceptibility (Feyisa et al., 2014).

SPECTRAL INDICES	FORMULA
Automated Water Extraction Index (AWEI)	$AWEI = 4 * (B3 - B11) - (0.25 * B8 + 2.75 * B12)$
Normalized Difference Vegetation Index (NDVI)	$NDVI = \frac{B8 - B4}{B8 + B4}$
Modified Normalized Difference Water Index (MNDWI)	$MNDWI = \frac{B3 - B11}{B3 + B11}$

Where:

- B3 (Green)
- B4 (Red)
- B8 (Near Infrared, NIR)
- B11 (Shortwave Infrared 1, SWIR-1)
- B12 (Shortwave Infrared 2, SWIR-2)

This dataset was therefore constructed by integrating static variables (DEM, HAND, TWI, curvature, and slope) and dynamic variables derived from 13 flood events (between October 2019 and August 2025) on which flood events were detected by the UN OCHA HDX platform. Of these dates, the first ten flood events in chronological order were used for training, while the three most recent flood events were reserved for testing. For each date, three spectral indices were derived: MNDWI, NDVI, and AWEI.

$$\left\{ \underbrace{dem, slope, hand, twi, curv}_{static}, \underbrace{point_pattern, vh, vv, ndvi, mndwi, awei}_{dynamic} \right\}$$

Both raw and processed data are available, resulting in a complete dataset for South Sudan designed to train flood detection models. To the best of our knowledge, this is the first public dataset of its kind in the academic literature. The dataset covers 10 flood-prone areas of interest and comprises 13 flood events per area, yielding a total of 130 flood rasters. Each observation includes seven bands combining static and dynamic variables. This dataset represents one of the main contributions of this work and directly addresses the research gap identified in the Introduction.

3. METHODOLOGY

3.1 METHODOLOGICAL OVERVIEW

The proposed methodology provides a probabilistic alternative to conventional artificial neural networks, while explicitly accounting for spatial dependence, uncertainty propagation, and physical interpretability. The modeling pipeline shown in diagram below (FIG 7) integrates both static and dynamic geospatial covariates as inputs explained in last section.

As reviewed in the Introduction, the core of the methodology lies a latent Gaussian random field defined over the spatial domain. This latent field is constructed through the SPDE formulation, which establishes a direct link between Gaussian random fields and Gaussian Markov random fields via a finite element discretization over a triangular mesh. The mesh defines the spatial support of the latent process and allows efficient approximation of continuous spatial variability while maintaining computational tractability (Lindgren et al., 2011).

Inference is performed using INLA, which provides accurate deterministic approximations to the posterior distributions of both latent spatial effects and model parameters without resorting to computationally expensive Markov Chain Monte Carlo methods. Within this framework, flood occurrence is modeled as a binomial probabilistic response variable, linked to covariates and the latent spatial field through a logit link function. The resulting output is a spatially explicit flood probability surface, estimated at the pixel level.

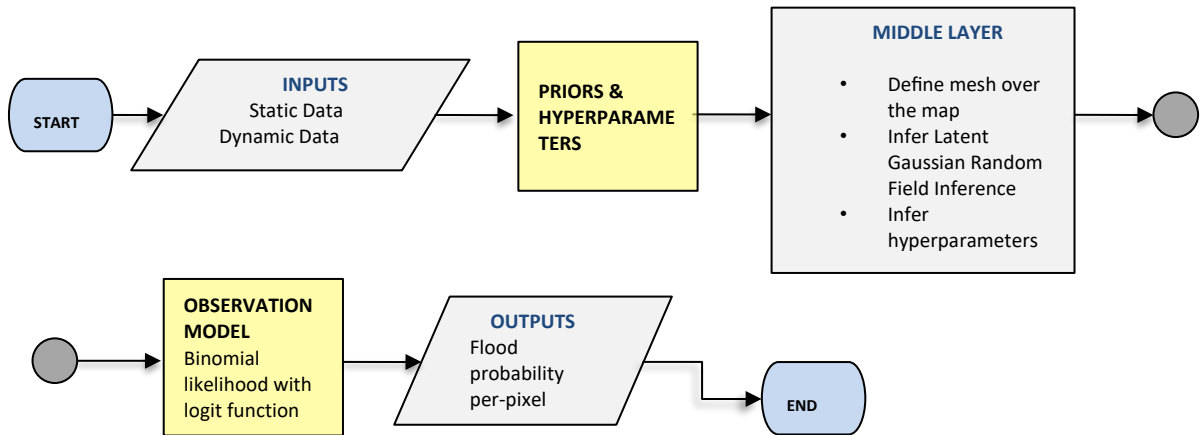


FIG 7. Overview of the INLA–SPDE proposed methodology for pixel-level flood probability estimation. Source: Author’s own elaboration

The reviewed methodology is implemented following an algorithmic framework.

3.2 METHODOLOGY. BAYESIAN FLOOD MAPPING VIA INLA-SPDE

INPUTS

- S_{train} : Multi-temporal training point pattern with binary flood indicators $y_i \in \{0,1\}$
- $X_{dynamic}$: Time-series of satellite rasters (S1-VV, S1-VH, NDVI, MNDWI, AWEI)
- X_{static} : Static topographic rasters (DEM, Slope, HAND, Curvature, TWI)

- $\mathcal{G}$: Study area of interest boundary and coordinate reference system (WGS 84 / UTM zone 36N - EPSG:32636)

OUTPUTS

- $\hat{p}(s)$: Continuous posterior flood probability map
- $\xi(s)$: Latent spatial Gaussian field (spatial structured uncertainty)
- Θ : Posterior distributions of fixed effects β and hyperparameters θ

STEPS METHODOLOGY

1. Data Harmonization and Feature Engineering

The model begins by aligning all multi-sensor inputs to a common spatial grid defined by a master reference.

- **Temporal Aggregation:** Dynamic predictors are aggregated via temporal averaging to capture the persistent spectral signature of the study period

$$\bar{X}_k = \frac{1}{T} \sum_{t=1}^T X_{k,t} \in \{vh, vv, ndvi, mndwi, awei\} \quad (9)$$

- **Standardization:** To ensure numerical stability and comparable coefficient scales, all covariates are transformed using Z-score normalization

$$z_{ik} = \frac{(x_{ik} - \mu_k)}{\sigma_k} \quad (10)$$

MIDDLE LAYER

2. Domain Discretization (Mesh Construction)

To transform the continuous spatial process into a computationally feasible Gaussian Markov Random Field (GMRF), a constrained triangulation is performed.

- A triangular mesh is generated over the study domain $\mathcal{D}$
- The mesh utilizes a multi-resolution approach: a fine inner zone for the observation area to minimize discretization error, and a coarser outer buffer (offset) to mitigate boundary effects (see figure below)

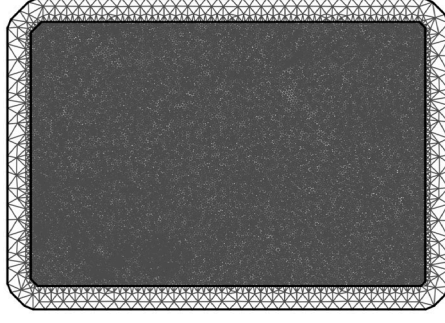


FIG 8. 2D triangular mesh over study domain $\mathcal{D}$. Source: Author's own elaboration

3. Spatial Stochastic Process (SPDE)

The latent spatial effect $f(s)$ is modeled as a solution to the following linear fractional Stochastic Partial Differential Equation: (for more detail, see Introduction)

$$(\kappa^2 - \Delta)^{\frac{\alpha}{2}} f(s) = W(s) \quad (11)$$

where Δ is the Laplacian operator, $W(s)$ is spatial white noise, and α determines the smoothness of the Matérn covariance. This link allows the model to treat the continuous field as a sparse GMRF, significantly reducing the computational cost of the covariance matrix inversion

OBSERVATION MODEL

4. Model Specification (Latent Gaussian Field)

The flood occurrence is modeled using a spatial logistic regression within a Bayesian framework

- Likelihood $y_i | \eta_i \sim \text{Bernoulli}(p_i) \equiv y_i | \eta_i \sim \text{Binomial}(n = 1, p_i)$
- Link Function $\text{logit}(p_i) = \eta_i$
- Linear predictor (equation 9)

$$\eta_i = \beta_0 + \sum_{k=1}^K \beta_k z_{ik} + f(s_i) \quad (12)$$

where:

β_k represents fixed effects, capturing linear feature learning from observed covariates, $f(s_i)$ is the spatial random effect over spatial locations s_i

- Priors: Penalized Complexity (PC) priors are assigned to the SPDE hyperparameters (range and marginal variance) to ensure sensible spatial correlation lengths relative to the study area diagonal

5. Bayesian Inference via INLA

Instead of using computationally intensive MCMC, the model employs Integrated Nested Laplace Approximation (INLA).

- The algorithm computes posterior marginals for the fixed effects and the latent field by using Laplace approximations for the integrals
- The strategy utilizes an Empirical Bayes (EB) integration scheme for the hyperparameters to accelerate convergence.

6. Spatial Prediction and Validation

The posterior mean of the linear predictor is projected from the mesh nodes to the full raster grid:

- Projector Matrix (A): Maps the GMRF from the triangulation vertices to the pixel coordinates
- Inverse Logit: Converts the latent predictor back to probability space $p = \frac{1}{1+e^{-\eta}}$
- Validation: Model performance is assessed using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) and the Conditional Predictive Ordinate (CPO)

3.3 ALGORITHM IMPLEMENTATION. OF BAYESIAN SPATIOTEMPORAL FLOOD MAPPING VIA INLA-SPDE

Phase 1: Data Alignment and Standardization

1. Spatial Harmonization: Project and resample all covariates X_{Static} and X_{dynamic} to a common master reference grid (Sentinel-1 VV base) using bilinear interpolation.
2. Temporal Aggregation: Calculate the temporal mean of the dynamic covariates across the observation period to establish baseline spatial conditions.
3. Standardization: Normalize the aggregated covariate stack using a Z-score transformation to ensure computational stability and comparable effect sizes (β coefficients).
4. Response Extraction: Spatially intersect the transformed raster stack with the binary observation points Y to construct the training matrix.

Phase 2: Spatial Domain Discretization (Mesh Construction)

5. Domain Definition: Extract the spatial coordinates $s_i = (x_i, y_i)$ from the observation points.
6. Construct a constrained 2D triangulated mesh over the spatial domain to discretize the continuous latent field
 - a. Inner Domain: Define a dense triangulation (max edge = diagonal/150) to capture fine-scale spatial variations in the region of interest.
 - b. Outer Extension: Define a coarser boundary (max edge = diagonal/20) to prevent boundary effect artifacts in the variance approximations.

Phase 3: SPDE Formulation and Matérn Covariance

7. Latent Field Definition: Model the spatial dependency as a continuously indexed Gaussian field $f(s)$ with a Matérn covariance structure.
8. Implicit Parameters (R-INLA Defaults): Approximate $f(s)$ by solving the Stochastic Partial Differential Equation (SPDE) from equation (5) (see Introduction).
9. Set the fractional operator $\alpha=2$ (default in `inla.spde2.pcmatern`)
10. For a 2D spatial domain ($d=2$), the Matérn smoothness parameter is implicitly defined as $v=\alpha-d/2=1$.
11. Prior Specification (PC Priors): Enforce Penalized Complexity priors to shrink the model towards a base model (no spatial effect):

- a. $P(\rho < \rho_0) = 0.01$ where ρ_0 is 20% of the domain's bounding box diagonal
- b. $P(\sigma < \sigma_0) = 0.01$ where $\sigma_0 = 1$

Phase 4: Constructing the Observation Matrix (The Stack)

- 12. Projection Matrix (A): Create a sparse projector matrix A that maps the continuously defined Gaussian Markov Random Field (GMRF) from the mesh nodes to the exact continuous observation locations s_i .
- 13. Linear Predictor: Formulate the spatial logistic regression model (see equation (12)).

Phase 5: Approximate Bayesian Inference via INLA

- 14. Likelihood: Assign a Binomial (Bernoulli) likelihood to the observations.
- 15. Model Fitting: Execute the Integrated Nested Laplace Approximation (INLA) to estimate the posterior marginals for $\theta = \{\beta, \rho, \sigma\}$ without Markov Chain Monte Carlo (MCMC) sampling.
 - a. Apply an adaptive step-size strategy and Empirical Bayes (eb) integration for the hyperparameters.
 - b. Disable Variational Bayes correction (`control.vb = FALSE`) to stabilize convergence during the Hessian evaluations.

Phase 6: Continuous Spatial Prediction (Testing)

- 16. Extraction: Extract the posterior means of the fixed effects and the spatial latent field.
- 17. Block Prediction: For each novel time step t in the test set:
 - a. Load and standardize the dynamic raster variables using the temporal baseline statistics
- 18. Divide the pixel grid into manageable chunks to optimize memory.
- 19. For each pixel u , compute the linear predictor (equation 12).
- 20. Apply the inverse logit link function to map values to the $[0,1]$ probability space:

$$\hat{p}_u = \frac{1}{1 + e^{(-\hat{\eta}_u)}}$$

- 21. Output: Assemble and export the resultant raster of $\hat{p}_u$ as the flood susceptibility map.

The proposed algorithm was implemented in the R programming environment, primarily using the [INLA package](#) to perform Bayesian inference under the Stochastic Partial Differential Equation (SPDE) framework. Spatial data preprocessing and orchestration, including raster harmonization, coordinate handling, and mesh construction, were carried out using the `sf`, `terra`, and `sp` packages. Model outputs, including flood susceptibility maps and posterior summaries, were visualized using `ggplot2` (see Results section). The predictive performance of the spatial logistic regression model was quantitatively assessed through the Area Under the Receiver Operating Characteristic Curve (AUC), computed using the `pROC package`. Overall, this computational framework enables an efficient, reproducible, and statistically robust estimation of latent Gaussian fields for multi-sensor flood mapping applications.

3.4 REPRODUCIBILITY AND SOFTWARE IMPLEMENTATION

Algorithm in last subsection is implemented in R. The complete source code is openly available and archived on Zenodo, with ongoing development hosted on GitHub (Lozada-Aguilar, 2026). The source code supporting the findings of this thesis is archived on Zenodo with a permanent DOI². The actively maintained development repository is hosted on GitHub at <https://github.com/cibernetista/INLA-SPDE-Bayesian-Flood-Mapping>

² <https://doi.org/10.5281/zenodo.18442227>

4. RESULTS AND DISCUSSION

4.1 POSTERIOR ESTIMATES OF FIXED EFFECTS

The Bayesian spatial logistic regression yielded posterior distributions for the ten geospatial and spectral predictors. Table 1 summarizes the mean effects, standard deviations, and 95% credible intervals (CI).

Table 1. Beta effects of the covariates

Covariate	mean	sd	0.025quant	0.5quant	0.975quant	mode	kld
Intercept	▼ - 9.43	▲ 0.10	▼ - 9.63	▼ - 9.43	▼ - 9.22	▼ - 9.43	○ -
dem	▼ - 0.06	▲ 0.02	▼ - 0.09	▼ - 0.06	▼ - 0.04	▼ - 0.06	○ -
slope	▼ - 0.06	▲ 0.01	▼ - 0.08	▼ - 0.06	▼ - 0.05	▼ - 0.06	○ -
hand	▼ - 0.09	▲ 0.01	▼ - 0.12	▼ - 0.09	▼ - 0.07	▼ - 0.09	○ -
curv	▼ - 0.00	▲ 0.01	▼ - 0.01	▼ - 0.00	▲ 0.01	▼ - 0.00	○ -
twi	▲ 0.01	▲ 0.01	▼ - 0.00	▲ 0.01	▲ 0.02	▲ 0.01	○ -
wv	▼ - 2.69	▲ 0.02	▼ - 2.73	▼ - 2.69	▼ - 2.65	▼ - 2.69	○ -
vh	▼ - 1.81	▲ 0.02	▼ - 1.85	▼ - 1.81	▼ - 1.78	▼ - 1.81	○ -
ndvi	▼ - 0.16	▲ 0.01	▼ - 0.17	▼ - 0.16	▼ - 0.14	▼ - 0.16	○ -
mndwi	▼ - 0.18	▲ 0.01	▼ - 0.21	▼ - 0.18	▼ - 0.15	▼ - 0.18	○ -
awei	▼ - 0.05	▲ 0.01	▼ - 0.08	▼ - 0.05	▼ - 0.03	▼ - 0.05	○ -

The estimated regression coefficients reveal physically consistent relationships between terrain, remote sensing variables, and flood occurrence. Negative and statistically significant coefficients for elevation (DEM) and slope confirm that low-lying and flat areas are more prone to water accumulation, while HAND emerges as the strongest topographic predictor, highlighting the vulnerability of areas closer to the drainage network.

The Topographic Wetness Index (TWI) is the only physical variable with a positive coefficient, indicating that higher terrain-driven moisture accumulation increases flood probability. Sentinel-1 radar backscatter variables (VV and VH) exhibit the largest negative effects, reflecting the well-known specular behavior of open water surfaces that strongly reduce returned radar energy. NDVI also shows a significant negative relationship, consistent with the absence of healthy vegetation in inundated or channelized areas. In contrast, the negative coefficients observed for optical water indices (MNDWI and AWEI) likely arise from multicollinearity with radar predictors and from mixed spectral responses in turbid or vegetated water bodies.

Finally, the very low intercept indicates that flooding is a spatially rare event within the study area, while the narrow credible intervals across predictors suggest high model confidence in the estimated effects.

4.2 LATENT SPATIAL PROCESS (SPDE)

The latent Gaussian field $f(s_i)$ captures the spatial autocorrelation not explained by the covariates. This process effectively regularizes the model and accounts for unobserved environmental factors. In other words, this map shows the remaining spatial trends, areas with high positive values in the latent field indicate areas where flooding occurred that indices could not explain on their own (e.g., due to signal saturation or previous cloud cover; see FIG 9 below).

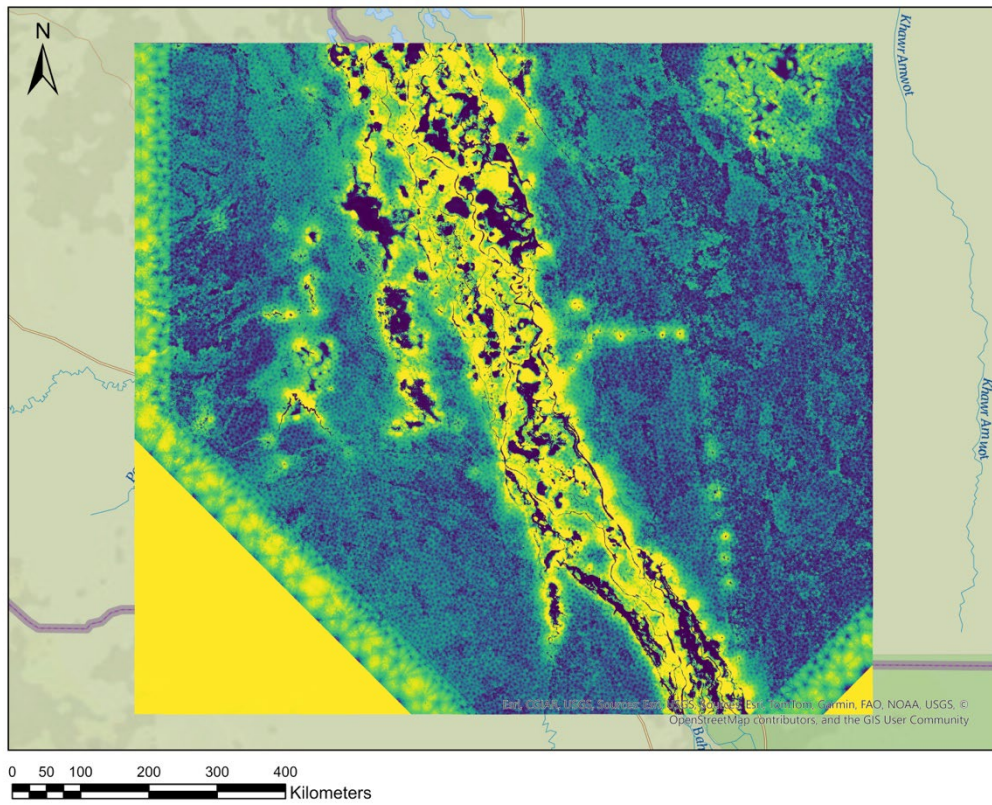


FIG 9. Spatial Latent Field (Gaussian field $f(s_i)$). Source: Author's own elaboration

Predictive Performance and Validation

The achieved AUC of 0.9949 indicates an exceptional ability of the model to discriminate between flooded and non-flooded pixels within the training domain. This high level of accuracy is attributed to the multi-sensor fusion strategy, which combines the all-weather penetration of SAR with the physical context provided by topographic indices. The near-perfect classification reflected in the Receiver Operating Characteristic (ROC) curve suggests that the selected predictors provide a clear and robust signal for the Bayesian classifier (see FIG 10).

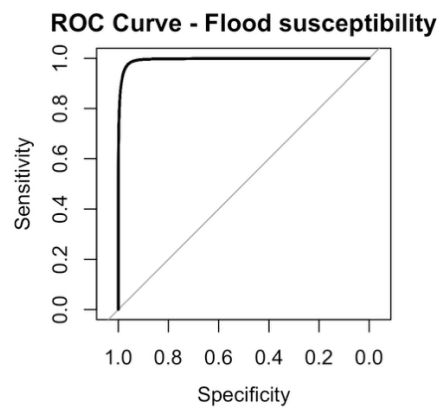


FIG 10. ROC Curve for selected predictors. X-Axis represents the true negative rate in the training process. Y-axis represents the true positive rate in the training process or model's ability to correctly identify flooded pixels. Source: Author's own elaboration

Despite the high predictive performance ($AUC = 0.9949$), this result should be interpreted with caution due to inherent limitations in spatial predictive modeling. In geospatial contexts, observations are typically not independent but spatially autocorrelated, meaning that nearby pixels tend to share similar characteristics. Consequently, if training and test samples are spatially proximate, belong to the same flood event, or share similar environmental covariates, the model may benefit from implicit spatial information leakage. Therefore, although the reported AUC indicates discrimination capability, it may overestimate the true generalization performance under fully independent spatial conditions.

4.3 FLOOD SUSCEPTIBILITY MAPPING

The final output is a continuous spatial probability surface, $\hat{p}(s)$, which integrates spectral evidence derived from multi-temporal remote sensing data with explicit spatial dependence captured by the model (see FIG 11 below).

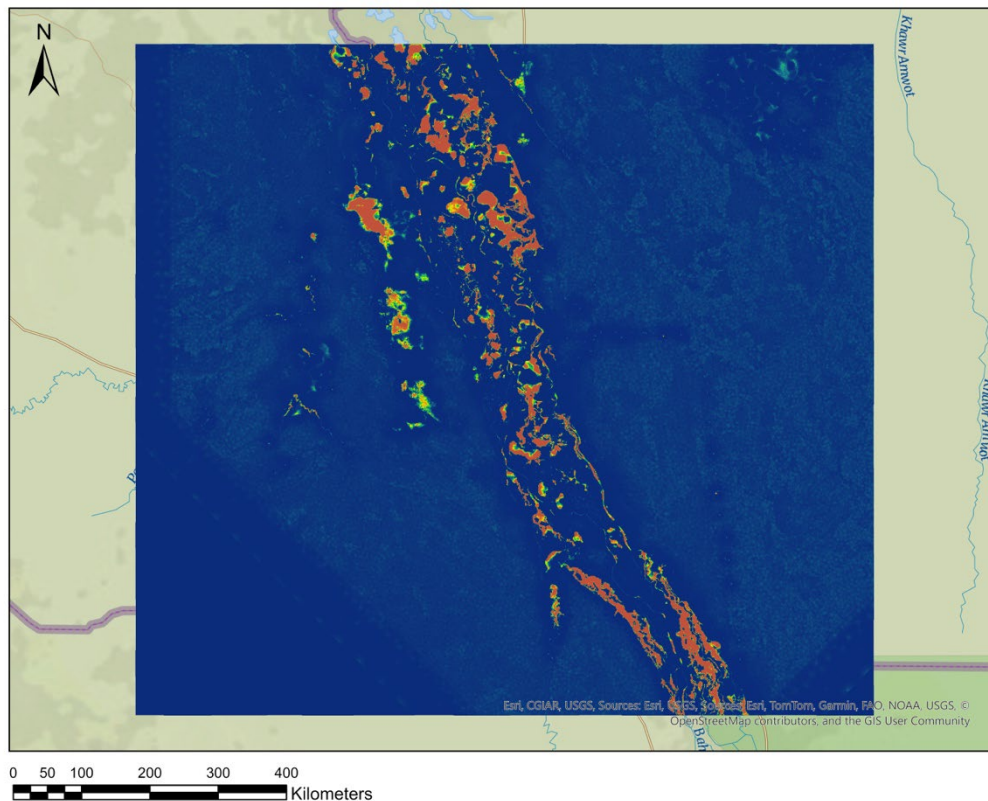


FIG 11. Final output. Flood probability per pixel. Source: Author's own elaboration

Unlike rigid threshold-based approaches applied to a single radar band, this probabilistic map represents a continuous gradient of flood susceptibility. Areas exhibiting high predicted probabilities (e.g., red-yellow tones) correspond to locations where low HAND values converge with elevated surface moisture signals identified across the multitemporal data stack, reflecting the combined influence of topographic control and hydrological conditions.

4.4 SPATIOTEMPORAL PREDICTION (TESTING STAGE)

The model was applied to unseen test dates to evaluate its robustness across different temporal snapshots. This experiment explicitly assesses the model's capacity to generalize beyond the training period. By applying the scaling statistics derived from the training data to the test observations, data leakage is avoided and consistency in feature representation is preserved, ensuring that the learned relationships remain

transferable to new acquisition dates, or, by extension, to comparable sensors, within the same geographic domain. See table below

Table 2. Performance metrics derived from the confusion matrix computed on the testing dataset, comprising three flood events not observed during the training phase

METRIC	VALUE	INTUITION
% of actual flooded pixels predicted	80.29%	% captured of total actual flooded pixels
% of accuracy flooded pixels	70.48%	% of predictions that were true (TP proportion vs TP + TN)
False positive rate (%)	0.74%	% of non-flooded pixels that were predicted as flooded
% of flood lost	19.71%	Percentage of flooded pixels that were not predicted

5. CONCLUSION AND FUTURE PERSPECTIVES

This research successfully fulfilled its primary objectives, addressed the proposed research questions, and contributed to partially closing the identified research gap in flood detection and spatial risk modeling. The proposed Bayesian framework demonstrated robust performance for flood mapping, achieving a twelve-fold improvement in spatial resolution compared to the reference flood products provided by the UN OCHA HDX platform. Furthermore, temporal resolution was substantially enhanced, as the forthcoming Sentinel-1C mission will enable the acquisition of SAR imagery over South Sudan at an approximate revisit time of 12 days, allowing for more timely and consistent flood monitoring.

Beyond flood detection, this work opens new perspectives for the application of Bayesian spatial models based on the INLA framework to other natural hazards, such as droughts and hydrological extremes through satellite observations. A key future challenge lies in the development of non-separable spatiotemporal models capable of jointly capturing spatial dependence and temporal dynamics. The current approach focuses on spatial variability and does not explicitly incorporate temporal structures such as seasonality, recurrence, or event frequency, which are critical components of hydrological processes in climate-sensitive regions.

From a data perspective, this study presents, for the first time, a comprehensive and harmonized dataset for South Sudan that integrates multi-sensor satellite observations with spatial covariates relevant to flood susceptibility. This dataset provides a valuable benchmark for comparative analyses and methodological experimentation. Future work should explore how this dataset performs under alternative modeling paradigms and epistemological approaches, including artificial intelligence techniques such as convolutional neural networks, transformer-based architectures, and the fine-tuning of foundation models such as IBM TerraMind (released on April, 2025).

Finally, it is expected that the outcomes of this research will be of practical relevance for the United Nations Global Service Centre. Continued development, validation, and methodological extensions of this framework will be openly shared through GitHub, fostering transparency, reproducibility, and collaborative advancement in geospatial disaster analytics (for further details, please visit <https://github.com/cibernetista/INLA-SPDE-Bayesian-Flood-Mapping>).

EPILOGUE. ON THE ANTHROPOLOGY OF CLIMATE CHANGE

As mentioned in the introduction, climate change is currently unfolding within a human-induced phase commonly referred to as the Anthropocene. Official data indicate that this process is largely driven by the annual emission of approximately 52 billion tons of carbon dioxide into the atmosphere, originating from a wide range of human activities, including the energy sector, construction, livestock production, cooling and heating systems, among others (Gates, 2022).

However, this phenomenon cannot be attributed to humanity as a homogeneous whole. Across the world, there exist diverse ontologies and forms of otherness such as human groups whose ways of inhabiting the Earth differ substantially from the dominant industrial paradigm. Many of these communities maintain relational worldviews grounded in long-standing ecological knowledge, including animistic or totemic perspectives, in which elements of nature are understood as living, relational, and, in some cases, spiritually or genealogically connected to human life.

Local and Indigenous knowledge systems are increasingly at risk of disappearance. This loss is not solely due to the fact that such systems may struggle to account for rapidly changing environmental phenomena in the same explanatory terms as Western scientific frameworks, but also because climate change directly affects the material conditions that sustain these communities. Forced displacement, land abandonment, and transformations of traditional economic activities are becoming more frequent as extreme events intensify and ecosystems undergo degradation, undermining livelihoods that have persisted for generations.

This brief reflection is included to emphasize the importance of understanding the Earth as a dynamic system, one that does not only narrate its history through climatic data and physical indicators, but also through the experiences, memories, and practices of the people who inhabit it. Now more than ever, local knowledge can offer a more relational and compassionate perspective on the planet we call home.

From this viewpoint, archaeology and anthropology may also be understood as forms of engagement or activism. Consider, for instance, Tlaloc, the Mexica god of rain (or Chaac, in Maya cosmology) is both a giver of life and a creative force. In contemporary contexts, these figures can be reinterpreted symbolically, as phenomena associated with rainfall and water cycles are increasingly perceived not only as sources of sustenance and life, but also as drivers of destruction through flooding and extreme events as the ones studied in this thesis, a whole new narrative different to the original indigenous knowledge.

I allow myself to include this reflection because, ultimately, my intention is to contribute value through science and technical expertise, while remaining grounded in the same underlying motivation: humanitarian concern. Bridging quantitative environmental modeling with cultural understanding is, in this sense, not an accessory to scientific work, but an ethical extension of it.

REFERENCES

- Bauer-Marschallinger, B., Cao, S., Tupas, M. E., Roth, F., Navacchi, C., Melzer, T., Freeman, V., & Wagner, W. (2022). Satellite-Based Flood Mapping through Bayesian Inference from a Sentinel-1 SAR Dacube. *Remote Sensing*, 14(15). <https://doi.org/10.3390/rs14153673>
- Bereczky, M., Wieland, M., Krullikowski, C., Martinis, S., & Plank, S. (2022). Sentinel-1-Based Water and Flood Mapping: Benchmarking Convolutional Neural Networks Against an Operational Rule-Based Processing Chain. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 15, 2023–2036. <https://doi.org/10.1109/JSTARS.2022.3152127>
- Chen, Y., & Bruzzone, L. (2022). A Self-Supervised Approach to Pixel-Level Change Detection in Bi-Temporal RS Images. *IEEE Transactions on Geoscience and Remote Sensing*, 60, 1–11. <https://doi.org/10.1109/TGRS.2022.3203897>
- Chol, D., Hall, J., Wheeler, K., Bernhofen, M., Di Vittorio, C., & Strzepek, K. (2026). Geospatial Analysis of Population Exposure to Flooding in the Sudd Region, South Sudan. *Journal of Flood Risk Management*, 19. <https://doi.org/10.1111/jfr3.70168>
- Cian, F., Delgado Blasco, J. M., & Ivanescu, C. (2024). Improving rapid flood impact assessment: An enhanced multi-sensor approach including a new flood mapping method based on Sentinel-2 data. In *Journal of Environmental Management* (Vol. 369). <https://doi.org/10.1016/j.jenvman.2024.122326>
- Downs, B., Kettner, A. J., Chapman, B. D., Brakenridge, G. R., O'Brien, A. J., & Zuffada, C. (2023). Assessing the Relative Performance of GNSS-R Flood Extent Observations: Case Study in South Sudan. *IEEE Transactions on Geoscience and Remote Sensing*, 61, 1–13. <https://doi.org/10.1109/TGRS.2023.3237461>
- Drakonakis, G. I., Tsagkatakis, G., Fotiadou, K., & Tsakalides, P. (2022). OmbriaNet—Supervised Flood Mapping via Convolutional Neural Networks Using Multitemporal Sentinel-1 and Sentinel-2 Data

-
- Fusion. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 15, 2341–2356. <https://doi.org/10.1109/JSTARS.2022.3155559>
- Esri. (n.d.-a). *How curvature works—ArcGIS Pro | Documentation*. Retrieved <https://pro.arcgis.com/en/pro-app/3.4/tool-reference/spatial-analyst/how-curvature-works.htm>
- Esri. (n.d.-b). *How slope works—ArcGIS Pro | Documentation*. Retrieved <https://pro.arcgis.com/es/pro-app/latest/tool-reference/spatial-analyst/how-slope-works.htm>
- Feyisa, G. L., Meilby, H., Fensholt, R., & Proud, S. R. (2014). Automated Water Extraction Index: A new technique for surface water mapping using Landsat imagery. *Remote Sensing of Environment*, 140, 23–35. <https://doi.org/10.1016/j.rse.2013.08.029>
- Gates, B. (2022). *How to avoid a climate disaster: The solutions we have and the breakthroughs we need* (1st edn). Penguin.
- Hansen, M., Vejby, J., & Koch, J. (2025). Flood mapping using Sentinel-1 imagery with topographical and hydrological contextualization: Case study from Ribe, Denmark. *International Journal of Applied Earth Observation and Geoinformation*, 143, 104816. <https://doi.org/10.1016/j.jag.2025.104816>
- Intergovernmental Panel on Climate Change (IPCC) (Ed.). (2023). Water. In *Climate Change 2022 – Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (pp. 551–712). Cambridge University Press. Cambridge Core. <https://doi.org/10.1017/9781009325844.006>
- Jong, S. M. de, & Meer, F. van der. (2004). Remote sensing image analysis: Including the spatial domain. In *Remote sensing image analysis: Including the spatial domain* (1st ed. 2004.). Springer.
- Konapala, G., Kumar, S. V., & Ahmad, S. K. (2021). Exploring Sentinel-1 and Sentinel-2 diversity for flood inundation mapping using deep learning. *ISPRS Journal of Photogrammetry and Remote Sensing*, 180, 163–173. <https://doi.org/10.1016/j.isprsjprs.2021.08.016>
- Kridalukmana, R., Eridani, D., & Septiana, R. (2023). A Fuzzy-Bayesian-Based Method for River Flood Early Warning System. In *Proceeding—COMNETSAT 2023: IEEE International Conference on Communication, Networks and Satellite* (pp. 347–352). <https://doi.org/10.1109/COMNETSAT59769.2023.10420680>

-
- Lindgren, F., Rue, H., & Lindström, J. (2011). An Explicit Link between Gaussian Fields and Gaussian Markov Random Fields: The Stochastic Partial Differential Equation Approach. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 73(4), 423–498. <https://doi.org/10.1111/j.1467-9868.2011.00777.x>
- Loukili, Y., Lakhrissi, Y., Bourian, I., Hassine, L., Kourab, Z., Ali, T., Elaksher, A. F., & Chakraborty, R. (2025). Enhancing Flood Mapping Accuracy in North and West Africa Using Multi-Sensor Satellite Data and Machine Learning in Google Earth Engine. In *Earth Systems and Environment*. <https://doi.org/10.1007/s41748-025-00796-8>
- Lozada-Aguilar, F. J. (2026). *Multi-Sensor INLA-SPDE-Bayesian-Flood-Mapping: A United Nations Case Study* (Version V1.0) [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.18442227>
- Meyer, R. P., Sjøgaard, M. G., Schødt, M. P., Horion, S., & Prishchepov, A. V. (2022). EMERGENCY FLOOD MAPPING IN AUSTRALIA WITH SENTINEL-1 AND SENTINEL-2 SATELLITE IMAGERY; [CARTOGRAPHIE D'URGENCE DES INONDATIONS EN AUSTRALIE À PARTIR D'IMAGES SATELLITAIRES SENTINEL-1 ET SENTINEL-2]. In *BSGLg* (Vol. 78, Issue 1, pp. 123–138). <https://doi.org/10.25518/0770-7576.6653>
- Nobre, A. D., Cuartas, L. A., Hodnett, M., Rennó, C. D., Rodrigues, G., Silveira, A., Waterloo, M., & Saleska, S. (2011). Height Above the Nearest Drainage – a hydrologically relevant new terrain model. *Journal of Hydrology*, 404(1), 13–29. <https://doi.org/10.1016/j.jhydrol.2011.03.051>
- Qin, C.-Z., Zhu, A.-X., Pei, T., Li, B.-L., Scholten, T., Behrens, T., & Zhou, C.-H. (2011). An approach to computing topographic wetness index based on maximum downslope gradient. *Precision Agriculture*, 12(1), 32–43. <https://doi.org/10.1007/s11119-009-9152-y>
- Rue, H., Martino, S., & Chopin, N. (2009). Approximate Bayesian Inference for Latent Gaussian models by using Integrated Nested Laplace Approximations. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 71(2), 319–392. <https://doi.org/10.1111/j.1467-9868.2008.00700.x>
- Schaffhauser, P. (2022). Climate Change through the Metabolism of “Green” Capitalism. Considerations of Ecology According to Marx. *Antropología. Revista Interdisciplinaria Del Inah*, 6(13), 7–23.

Stephens, E. M., Bates, P. D., Freer, J. E., & Mason, D. C. (2012). The impact of uncertainty in satellite data on the assessment of flood inundation models. *Journal of Hydrology*, 414–415, 162–173. <https://doi.org/10.1016/j.jhydrol.2011.10.040>

Ta, L., Yu, C., Li, Z., Hu, X., Song, C., Huang, W., & Zhou, M. (2024). Dynamic flood mapping by a normalized probabilistic classification method using satellite radar amplitude time series. In *GIScience and Remote Sensing* (Vol. 61, Issue 1). <https://doi.org/10.1080/15481603.2024.2380125>

Zandsalimi, Z., Barbosa, S. A., Alemazkoor, N., Goodall, J. L., & Shafiee-Jood, M. (2025). Deep learning-based downscaling of global digital elevation models for enhanced urban flood modeling. *Journal of Hydrology*, 653, 132687. <https://doi.org/10.1016/j.jhydrol.2025.132687>





Masters Program in **Geospatial Technologies**



Supported by:



Education and Culture

ERASMUS MUNDUS